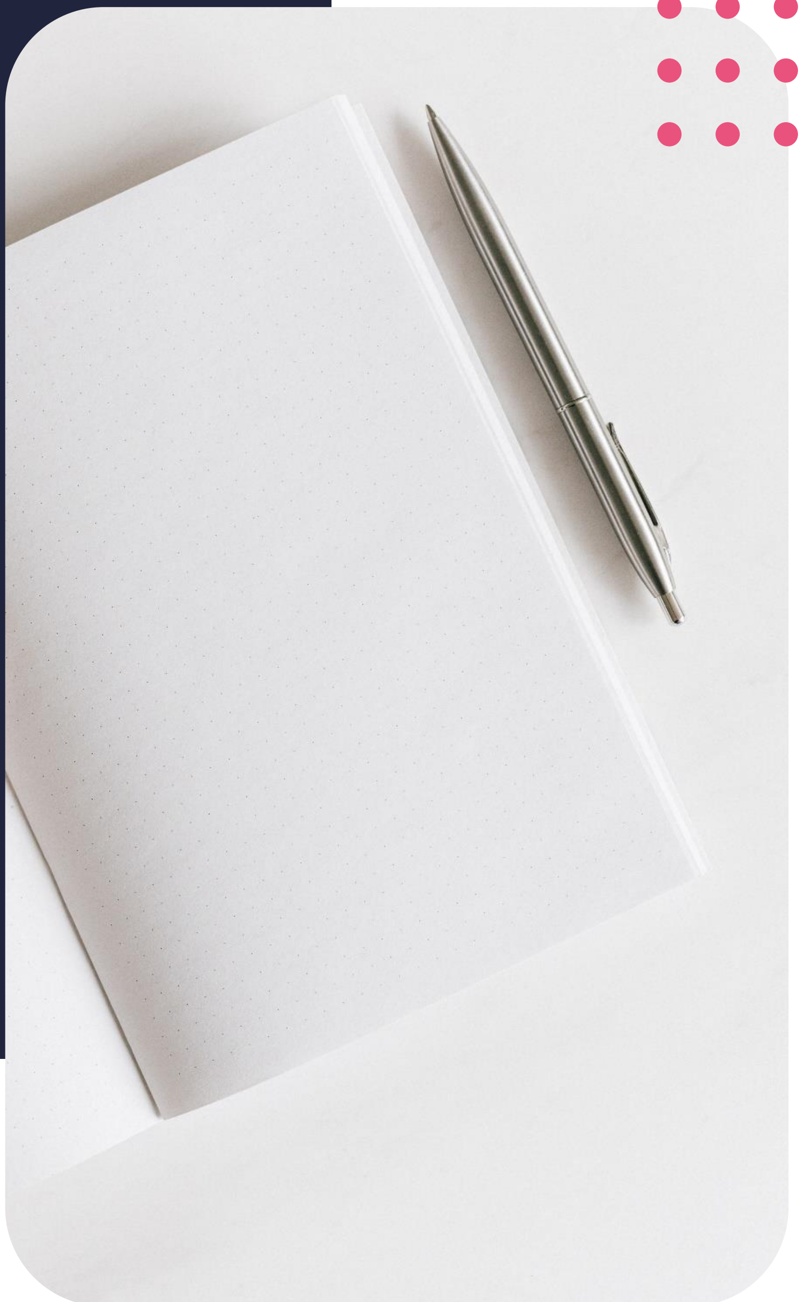
Diploma in Data Analytics

Linear regression continued

Contents



Lesson outcomes

By the end of this lesson, you should be able to:

- Linear regression continued
- Data frames continued
- Dates and times

Introduction

In this lesson we will continue to broaden our understanding of linear regression and data frames. We will understand what it means for the model to fit the data well and gain some further insight into treating data in R. We will end the lesson by exploring some basics surrounding dates values in R.

Linear regression continued

Linear regression recap

In the previous lesson, we learnt that a simple linear regression aims to model the relationship between the independent variables, our unknowns, and the dependent variable, our outcome variable, through fitting a straight line equation to the data. The model therefore assumes that the relationship between the predictor variable X and the response variable Y is linear.

$$\hat{y} = a + bx_i$$

Mathematically, we can write the linear regression relationship as the prediction of estimate of y that is represented by the intercept and the slope terms. These estimates are used to predict the value of the outcome variable.

We also learnt that we can use the function lm() to fit a simple linear regression model in R.

The function lm.fit provides us with some basic information about the model and summary(lm.fit) provides us with more detailed information.

Multiple linear regression

Single linear regression is useful when we only have one predictor variable to a response or outcome variable, but in practise, this is often not the case. We use a multiple linear regression model to fit multiple predictors to a single outcome variable.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \varepsilon$$

The mathematical model now contains multiple coefficients and unknown variables. Multiple linear regression therefore tries to model the relationship between 2 or more predictor variables and the response variable through fitting a linear equation to the data.

Model fit statistics

Residual standard error

- In a simple linear model, the residual standard error (RSE) represents an estimate of the standard deviation of the random error term.
- Error terms are associated with each observation, because we are never able to perfectly predict Y from our given set of observations from a sample.
- The RSE statistic therefore tells us more about the average amount that the response variable will stray from the true regression line.
- This value tells us how badly the current model fits the data.
- If the RSE value is small, the model fits the data well and vice versa if the RSE value is large.

R^2

- The r-squared statistic is also known as the coefficient of determination.
- The r-square statistic is used to explain the measure of variability in the response explained by the model in a simple linear regression model.
- With multiple linear regression, the r-squared statistic equals the square of the correlation between the response and the fitted linear model.
- An r-squared value near 1 show that the model explains a large part of the variance in the response variable.

Note: the r-squared value will always increase as we add more predictor variables to the model, even if the predictor variables added are not strongly associated with the response variable. We must therefore be careful not to add variables to the model that provides no real improvement to the model fit, but more on this in later lessons.

Another note: a large r-squared value does not necessarily mean that the estimated regression line fits the data well; another function, for example a polynomial trend, might describe the data better.

Data frames continued

tibble

By now you should be familiar with the collection of data analytics tool, tidyverse, that is used to transform and visualize data in R. Instead of using the traditional data frames that R mostly uses, tidyverse makes use of what it calls tibbles. These tibbles are actually data frames, but it has a few tweaks to it where Wickham thought the original concept of the data frame could use improvement. Tibbles make tweaks to data frame behaviour in order to make life working with them easier.

Apart from being described as a modern reimagining of data frames in R, tibbles is also called lazy and surly data frames because they do less than data frames and they complain more. The complaining the tibbles do, actually pressures us to tackle the problems in the data earlier which in the long run, makes our code cleaner and more efficient.

To name a few of the basic functions of tibbles:

- Using the tibble package in r, we are able to convert a data frame into a tibble with the function as_tibble.
- We can also create a new tibble using the function tibble()
- Furthermore, we are also able to define a tibble row by row with the function tribble(). Tribble is short for transposed tibble.

Tibble vs data frame

- Tibbles have a delightful printing method that show only the first 10 rows and all the columns that fit on the screen. This is handy when you work with large data sets.
- Subsetting a tibble will always return a tibble. You don't need to use *drop = FALSE* in contrast to conventional data.frames.

Dates and times

Lubridate package

R has a package called Lubridate that makes working with dates and times easier. Lubridate was a joint venture created by Hadley Wickham, the creator of Tidyverse, and Garrett Grolemund. It is currently being maintained by Vitalie Spinu.

Current date

- To get current timezone information, use command
  - Sys.timezone()
- To get current date, use command
  - Sys.date()
- To get current date, time and timezone particulars, use command
  - Sys.time()
- In Lubridate, to get current date, time and timezone particulars, use command
  - Now()

Converting strings to dates

Dates are often converted to character strings when imported into R.

- We can convert a string that is in date format to a date object with the use of the as.date function.
- The default date format for R is year-months-days.

Note: For our American friends, you will need to add an additional part to this format to convert the format of the date by telling R what format your date is in.

If using the Lubridate package,

- Use the command ymd or mdy depending on the format of the date to convert the string to a date object.

References

- Boehmke, B., 2020, *Dealing with Dates*, UC Business Analytics R Programming Guide, http://uc-r.github.io/dates/
- James, G., Witten, D., Hastie, T. & Tibshirani, R., 2013, *An Introduction to Statistical Learning with Application in R*, Springer, https://faculty.marshall.usc.edu/gareth-james/ISL/ISLR%20Seventh%20Printing.pdf
- Shekar, 2018, Model on Boston (simple linear regression), RPubs, https://rpubs.com/shekar07/399193
- Yue Xie, A., 2017, Regression with R – Boston Housing Price, Kaggle, https://www.kaggle.com/andyxie/regression-with-r-Boston-housing-price
- JMP Statistical Discovery from SAS, Fitting the Multiple Linear Regression Model, 2019, https://www.jmp.com/en_us/statistics-knowledge-portal/what-is-multiple-regression/fitting-multiple-regression-model.html
- Spinu, V., 2020, ymd, RDocumentation, https://www.rdocumentation.org/packages/Lubridate/versions/1.7.9/topics/ymd
- Spina, V., 2020, Package ‘lubridate’, R-Project, https://cran.r-project.org/web/packages/Lubridate/Lubridate.pdf
- Seth, K., 2020, EDA and Multiple Linear Regression on Boston Housing in R, Analytics Vidhya, https://medium.com/analytics-vidhya/eda-and-multiple-linear-regression-on-Boston-housing-in-r-270f858dc7b